NUMERICAL SIMULATION OF TRANSIENT VIBRATIONAL POWER FLOWS IN SLENDER RANDOM STRUCTURES

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This paper deals with some recent developments for the modeling and numerical simulation of high-frequency (HF) vibrations of randomly heterogeneous slender structures. The mathematical model is derived from the semiclassical analysis of strongly oscillating (HF) solutions of quantum and classical wave systems, including acoustic, electromagnetic, or in the present case elastic waves. This theory shows that the associated phase-space energy density satisfies a radiative transfer equation in a random medium at length scales comparable to the small wavelength. The proposed model also considers energetic boundary and interface conditions consistent with the boundary and interface conditions imposed to the solutions of the wave system. They are given in the form of power flow reflection/transmission operators for the energy rays impinging on a boundary or an interface. Specular-like transverse boundary reflections, diffuse reflections, or fluid-structure coupling may be treated as a particular case of the proposed model. Nodal/spectral discontinuous 'Galerkin' finite element methods and Monte-Carlo methods are implemented to integrate the radiative transfer equations with such boundary and interface conditions. Some numerical simulations are presented to illustrate the theory: the first one deals with an assembly of random thick beams, and the second one with an assembly of random thick shells. This research applies to the prediction of the linear transient responses of complex structures to impact loads or shocks, or the analysis of non-destructive evaluation techniques.

Keywords: High-frequency, vibration, radiative transfer, discontinuous finite elements, Monte-Carlo method.

1. Introduction

Engineering structures exhibit typical transport and diffusive behaviors in the higher frequency (HF) range of vibration (Savin, 2002). These regimes are described by linear transport equations for the energy density associated to the strongly oscillating (HF) solutions of the Navier-Cauchy equation for elastic wave propagation. More generally this result holds for all symmetric hyperbolic systems, including quantum waves and classical acoustic or electromagnetic waves (Gérard *et al.*, 1997, Guo & Wang, 1999, Papanicolaou & Ryzhik, 1999, Erdös & Yau, 2000, Bal, 2005, Powell & Vanneste, 2005, Lukkarinen & Spohn, 2007). It has been specialized to slender visco-elastic structures, typically beams, plates and shells, and fluid-saturated poro-visco-elastic media in Savin (2004, 2005a,b, 2007) for applications to aerospace structures. The mathematical model is derived from the semiclassical analysis of HF solutions of such wave systems: it shows that the associated energy density in the phase-space position $\times$ wave vector satisfies a radiative transfer equation (RTE) in a random medium at length scales comparable to the small wavelength. The RTE allows to track the energy paths

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(rays) within the propagation medium, thus it also describes all power flows in slender structures. This theory provides a rational framework for the validation, and generalization, of heuristic methods such as the statistical energy analysis (SEA, see Lyon & DeJong (1995)) or the vibrational conductivity analogy (VCA, see Nefske & Sung (1989)). Its main difficulty to date lies in the consideration of boundary and interface conditions for quadratic quantities, e.g. the energy and power flow densities, consistent with the boundary and interface conditions imposed to the displacement and stress fields of the original elastic wave system. The former are given as power flow reflection/transmission operators for the energy rays impinging on a boundary or an interface. Specular-like transverse boundary reflections, diffuse reflections, or fluid-structure coupling may be treated as a particular case. A formal derivation of these HF operators at interfaces between slender substructures such as beams or shells has been proposed in Savin (2007). More recently, a systematic derivation of the elastic energy density boundary conditions near the doubly-hyperbolic set (where both compressional and shear elastic waves are incident on the boundary) and the hyperbolic-elliptic set (where shear waves are incident on the boundary with an angle greater than their critical angle) has been proposed in Akian (2010). The analysis however does not account for the glancing set (at critical incidence) and the transport of energy along an interface by so-called gliding rays; this issue is the subject of ongoing research. Yet a radiative transfer model in bounded media with general transverse or diffuse boundary conditions for the power flows may be derived from these results (Savin, 2011).

RTEs can be solved numerically by various methods (Duderstadt & Martin, 1979, Dautray & Lions, 1993), including Monte-Carlo methods (Lapeyre *et al.*, 2003) or nodal/spectral discontinuous finite element methods (Reed & Hill, 1973, Lesaint & Raviart, 1974, Cockburn *et al.*, 2000, Brezzi *et al.*, 2004, Hesthaven & Warburton, 2008). Both techniques offer the great advantage of being numerically weakly dissipative and weakly dispersive (depending on the order of interpolation with finite elements). Thus they are well suited for long-time simulations, when RTEs converge to diffusion equations (Dautray & Lions, 1993, Papanicolaou & Ryzhik, 1999, Savin, 2008). The diffusive regime is characterized by energy equilibration rules which can be exhibited from direct numerical simulations using those schemes (Savin, 2011). Its onset at late times is prompted by the multiple scattering of energy rays on random heterogeneities in the medium–as modelized by the RTEs themselves–and to a lesser extent on boundaries and interfaces. It should be noted that the corresponding reflection/transmission operators need not be diffusive in order to obtain diffuse waves and rays within the medium itself; see for example Savin (2011) and many applications in architectural acoustics, among others. The diffusive regime is precisely the one invoked in the SEA and VCA approaches to derive steady-state global or local diffusion equation by heuristic arguments. The main purpose of this paper is to outline the radiative transfer theory actually yielding diffusion for high-frequency structural vibrations, and more specifically the numerical methods which may be used to solve transient RTEs in view of demonstrating this diffusive behavior at late times. It is organised as follows. First, the transport model for the evolution of the high-frequency energy density in randomly heterogeneous slender structures is resumed, considering boundary and interface conditions as well. The numerical methods implemented for solving the corresponding RTEs are detailed in Sect. 3. Then two numerical examples are introduced in Sect. 4, which resumes and extends the numerical simulation of transient transport in an assembly of random thick beams depicted in Savin (2011) and presents some new simulations of multi-group transport in an assembly of random cylindrical shells. At last Sect. 5 offers some conclusions and directions of future works.

2. Transport model

In Sect. 2.1 one summarizes the basic results obtained in Guo & Wang (1999), Papanicolaou & Ryzhik (1999), Bal (2005) for the transport properties of high-frequency waves in random elastic media. The boundary conditions to be added to the evolution model recalled below are detailed in Sect. 2.2 for specular-like or diffusive-like reflections at outer boundaries and in Sect. 2.3 for reflections and transmissions at interfaces.

2.1. *Radiative transfer in an open domain*

Let us consider an heterogeneous, viscoelastic medium embedded in the open domain $\mathcal{O} \subseteq \mathbb{R}^d$, where $\mathbb{R}$ is the set of all real numbers and $d = 1, 2$ or 3 is the physical space dimension. It is assumed that this medium exhibits statistically isotropic random perturbations of its mechanical parameters, such as the density and the Young or shear modulus. Then the evolution of the high-frequency energy density associated to high-frequency waves propagating in $\mathcal{O}$ follows the multigroup radiative transfer equations (MRTE) (Guo & Wang, 1999, Papanicolaou & Ryzhik, 1999):

$$\partial_t w_\alpha + \{\lambda_\alpha, w_\alpha\} + \Sigma_\alpha w_\alpha = \sum_{\beta=1}^{M} \int_{\mathbb{S}^{d-1}} \sigma_{\alpha\beta}(\mathbf{x}, \hat{\mathbf{k}} \cdot \hat{\mathbf{k}}') w_\beta(\mathbf{x}, \mathbf{k}', t)\, \mathrm{d}\mu(\hat{\mathbf{k}}')\,, \tag{1}$$

which couple M different propagation modes, or energy rays, of intensity w_α for $1 \leq \alpha \leq M$. Typically $M = 3$ and $\alpha = \mathrm{P}$, SV or SH in an isotropic elastic medium, for the compressional and shear waves. Here $\lambda_\alpha(\mathbf{x}, \mathbf{k}) = c_\alpha(\mathbf{x})|\mathbf{k}|$ is the eigenfrequency of the mode α of which group velocity is c_α, $\{f, g\} = \boldsymbol{\nabla}_{\mathbf{k}} f \cdot \boldsymbol{\nabla}_{\mathbf{x}} g - \boldsymbol{\nabla}_{\mathbf{x}} f \cdot \boldsymbol{\nabla}_{\mathbf{k}} g$ is the usual Poisson's bracket, $\mathbb{S}^{d-1}$ is the unit sphere of $\mathbb{R}^d$ with the uniform probability measure μ, and $\sigma_{\alpha\beta} \geq 0$ is the scattering cross-section which gives the rate of conversion of an energy ray β in the direction $\hat{\mathbf{k}}'$ to another ray α in the direction $\hat{\mathbf{k}}$, at position $\mathbf{x}$ and wavenumber $|\mathbf{k}|$; the standard notation $\mathbf{k} = |\mathbf{k}|\hat{\mathbf{k}}$ is used throughout this paper. At last the total scattering cross-section $\Sigma_\alpha \geq 0$ for a mode α is defined by:

$$\Sigma_\alpha(\mathbf{x}, \mathbf{k}) = \sum_{\beta=1}^{M} \int_{\mathbb{S}^{d-1}} \sigma_{\alpha\beta}(\mathbf{x}, \hat{\mathbf{k}} \cdot \hat{\mathbf{k}}')\, \mathrm{d}\mu(\hat{\mathbf{k}}')\,. \tag{2}$$

In a deterministic medium all scattering cross-sections are zero and right-hand-sides vanish in Eq. (1), which reduces to Liouville transport equations. Here the $\sigma_{\alpha\beta}$'s are assumed to depend on the directions $\hat{\mathbf{k}}$ and $\hat{\mathbf{k}}'$ through their scalar product solely, yielding isotropic scattering, but more general scattering processes can be considered by which they depend on $\mathbf{k}$ and $\mathbf{k}'$. They are given explicitly as functions of the auto- and cross-power spectra of the random perturbations of the medium mechanical parameters (Guo & Wang, 1999, Papanicolaou & Ryzhik, 1999, Bal, 2005). Note finally that the scalar Wigner measures

$\{w_\alpha\}_{1\leq\alpha\leq M}$, hereafter called phase-space energy densities or specific intensities, are non-negative if the initial data are non-negative. Then the space-time energy density $\mathcal{E}(\mathbf{x},t) \in \mathbb{R}_+$ and power flow density $\mathbf{\Pi}(\mathbf{x},t) \in \mathbb{R}^d$ of this medium are recovered by:

$$\begin{aligned}\mathcal{E}(\mathbf{x},t) &\quad \frac{1}{2}\sum_{\alpha\quad 1}^{M}\int_{\mathbb{R}^d} w_\alpha(\mathbf{x},\mathbf{k},t)\,\mathrm{d}\mathbf{k}\,,\\ \mathbf{\Pi}(\mathbf{x},t) &\quad \frac{1}{2}\sum_{\alpha\quad 1}^{M} c_\alpha(\mathbf{x})\int_{\mathbb{R}^d} w_\alpha(\mathbf{x},\mathbf{k},t)\hat{\mathbf{k}}\,\mathrm{d}\mathbf{k}\,.\end{aligned} \tag{3}$$

These results are adapted to elastic waves in slender structures in Savin (2004, 2005b, 2007), but they also apply to waves in poro-visco-elastic media, acoustic waves, electromagnetic waves, or the Schrödinger equation in random media (Erdös & Yau, 2000, Powell & Vanneste, 2005, Savin, 2005a, Lukkarinen & Spohn, 2007).

For convenience, the MRTE of Eq. (1) are reshaped in the form of a system of first-order conservative equations (5) in phase space $T^*\mathcal{O} \;:\quad \mathcal{O}\times\mathbb{R}^d$ as follows. Let us introduce the phase variable $\mathbf{z} \quad (\mathbf{x},\mathbf{k}) \in T^*\mathcal{O}$, the set $E \quad \{1,2,\dots M\}$, the vector $\mathbf{W} \quad (w_1,w_2,\dots w_M)^{\mathrm{T}}$ of $(\mathbb{R}_+)^M$, and the linear flux operator:

$$\begin{aligned}\mathcal{F}(\mathbf{W}) &\quad (w_1\mathbf{F}_1, w_2\mathbf{F}_2,\dots w_M\mathbf{F}_M)^{\mathrm{T}}\,,\\ \mathbf{F}_\alpha(\mathbf{z}) &\quad \begin{pmatrix} c_\alpha(\mathbf{x})\hat{\mathbf{k}}\\ -|\mathbf{k}|\boldsymbol{\nabla}_{\mathbf{x}}c_\alpha\end{pmatrix}\end{aligned}$$

($\mathcal{F}(\mathbf{W})$ is thus a $M\times 2d$ real matrix). The so-called collision operator $\mathcal{Q}_{\mathbf{x}}$ in Eq. (1) is defined by:

$$\mathcal{Q}_{\mathbf{x}}(\mathbf{W}) \quad \mathcal{K}_{\mathbf{x}}(\mathbf{W}) - \mathbf{\Sigma}\mathbf{W}\,, \tag{4}$$

where $\mathbf{\Sigma}(\mathbf{x},\mathbf{k}) \quad \mathrm{diag}(\Sigma_\alpha(\mathbf{x},\mathbf{k}))_{\alpha\in E}$, and:

$$\mathcal{K}_{\mathbf{x}}(\mathbf{W})(\mathbf{k}) \quad \int_{\mathbb{S}^{d-1}} \sigma(\mathbf{x},\hat{\mathbf{k}}\cdot\hat{\mathbf{k}}')\;\mathbf{W}(\mathbf{k}')\mathrm{d}\mu(\hat{\mathbf{k}}')\,,$$

where $\sigma(\mathbf{x},\hat{\mathbf{k}}\cdot\hat{\mathbf{k}}')_{\alpha\beta} \quad \sigma_{\alpha\beta}(\mathbf{x},\hat{\mathbf{k}}\cdot\hat{\mathbf{k}}')$, $\alpha,\beta\in E$. Then Eq. (1) writes:

$$\partial_t\mathbf{W} + \mathbf{Div}_{\mathbf{z}}\mathcal{F}(\mathbf{W}) \quad \mathcal{Q}_{\mathbf{x}}(\mathbf{W})\,, \tag{5}$$

which is a conservative system in the sense that $\int_{\mathbb{R}^d}\mathcal{Q}_{\mathbf{x}}(\mathbf{W})(\mathbf{k})\mathrm{d}\mathbf{k} \quad \mathbf{0}$ whenever $\mathcal{Q}_{\mathbf{x}} \in L^1_{\mathbf{k}}(\mathbb{R}^d)$. On any discontinuity front Σ_D in phase- space defined by the hypersurface $\Sigma_D \quad \{\mathbf{z}\in T^*\mathcal{O};\, S(\mathbf{z}) \quad 0\}$, its piecewise continuous (weak) solution $\mathbf{W}$ satisfies the Rankine-Hugoniot condition:

$$\sum_{\alpha\in E}[\![\{S,\lambda_\alpha\}w_\alpha]\!] \quad 0\,,$$

where $[\![f]\!]$ stands for the jump of f on the front Σ_D. This relation states that the total normal power flow density $\sum_{\alpha\in E}c_\alpha w_\alpha(\hat{\mathbf{k}}\cdot\hat{\mathbf{n}}_\Sigma)$ remains continuous accross that front, denoting by $\hat{\mathbf{n}}_\Sigma$ its normal; in return the specific intensities themselves have no reason to remain continuous. The Rankine-Hugoniot condition does not describe how the individual normal power flows $c_\alpha w_\alpha\hat{\mathbf{k}}\cdot\hat{\mathbf{n}}_\Sigma$ for each mode $\alpha\in E$ are distributed inward and outward on the front. This issue is considered in the next two sections for Σ_D being either a boundary or an interface (a junction) where the group velocities in the medium are discontinuous.

2.2. *Radiative transfer in a bounded domain*

Now the above system (5), completed by square-integrable initial conditions $\mathbf{W}(\mathbf{z},0) \quad \mathbf{W}^0(\mathbf{z})$, is considered in the phase space $T^*\mathcal{D} \quad \mathcal{D}\times\mathbb{R}^d$ where $\mathcal{D}$ is a bounded subset of $\mathcal{O}$, and for $t\in \;0,T\;$, $T<+\infty$ but arbitrary. It is assumed that $\mathbf{W}^0(\mathbf{x},\{\mathbf{k}\quad\mathbf{0}\}) \quad \mathbf{0}$. Wavenumber $|\mathbf{k}|>0$ is a fixed parameter, and we shall also assume as usual that the velocities $c_\alpha\in W^{1,\infty}(\mathcal{D})$ are positive functions with possible isolated discontinuities if $\mathcal{D}$ contains sharp interfaces between different materials. The right-hand-side collision operator is continuous on $L^2(T^*\mathcal{D})$ provided that $\sigma_{\alpha\beta} \in L^2(\mathcal{D} \times \mathbb{S}^{d-1} \times \mathbb{S}^{d-1})$ and $\Sigma_\alpha \in L^\infty(T^*\mathcal{D})$, $\alpha, \beta \in E$. For any smooth manifold Σ of co-dimension 1 in $\mathbb{R}^d$ oriented by its normal $\hat{\mathbf{n}}_\Sigma$, let:

$$\Gamma^\pm(\Sigma) \quad \{(\mathbf{x}, \mathbf{k}) \in \Sigma \times \mathbb{R}^d;\ \pm\hat{\mathbf{k}} \cdot \hat{\mathbf{n}}_\Sigma(\mathbf{x}) > 0\}\,.$$

Then well-posedness of the problem is ensured by setting proper boundary conditions on the inflow boundary $\Gamma^-(\partial\mathcal{D})$ of the computational domain $\mathcal{D}$ (Bensoussan *et al.*, 1979). Denoting by $\hat{\mathbf{n}}$ the outward unit normal to the (smooth) boundary $\partial\mathcal{D}$, an incoming flux $\mathbf{f}(\mathbf{z}, t)$ can be imposed as $-\mathcal{F}(\mathbf{W})\hat{\mathbf{n}} \quad \mathbf{f}$ on $\Gamma^-(\partial\mathcal{D})$ and for $t > 0$. However generalized boundary reflection laws can also be considered in the form:

$$-\mathcal{F}(\mathbf{W})\hat{\mathbf{n}} \quad \mathbf{f} + \mathcal{R}_\mathbf{x}(\mathbf{W})\,, \tag{6}$$

$\mathbf{z} \in \Gamma^-(\partial\mathcal{D})$, $t > 0$, where $\mathcal{R}_\mathbf{x}$ is a linear boundary reflection operator defined from $\Gamma^+(\partial\mathcal{D})$ into $\Gamma^-(\partial\mathcal{D})$. Note that it depends only on the *local* geometry of the boundary (it may also depend on $|\mathbf{k}|$). This operator describes how the outward normal power flows $\mathcal{F}(\mathbf{W})\hat{\mathbf{n}}$, $\hat{\mathbf{k}} \cdot \hat{\mathbf{n}} > 0$, are reflected into inward normal power flows $-\mathcal{F}(\mathbf{W})\hat{\mathbf{n}}$, $\hat{\mathbf{k}} \cdot \hat{\mathbf{n}} < 0$, with possible mode conversions in E. Here the glancing set $\hat{\mathbf{k}} \cdot \hat{\mathbf{n}} \quad 0$ is ignored, not because such grazing rays are unlikely to occur, but because we lack a theoretical model to describe the associated boundary specific intensities at present. The rigourous construction of $\mathcal{R}_\mathbf{x}$ away from the glancing set is detailed in Akian (2010) for elastic waves. A formal derivation for bending, compressional and shear waves in thick beams and shells is given in Savin (2007). Its basic properties and other examples are considered in Savin (2011).

2.3. *Radiative transfer with a sharp interface*

Assume that the physical domain $\mathcal{D}$ is partitioned into non-overlapping sub-domains and their interface $\gamma_D \subset \overline{\mathcal{D}}$ is a smooth, bounded manifold of codimension 1 oriented by its unit normal $\hat{\mathbf{n}}_D$ (ignoring edges and corners at a first glance). More specifically, γ_D is any interface between these sub-domains or between a sub-domain and the exterior of $\mathcal{D}$ and consequently, it may also be a part or the whole of $\partial\mathcal{D}$. If the group velocities c_α are continuous across γ_D, the continuity of the normal power flow on γ_D is ensured by the conditions $\mathcal{F}(\mathbf{W}^+)\hat{\mathbf{n}}_D \quad \mathcal{F}(\mathbf{W}^-)\hat{\mathbf{n}}_D$ if $\hat{\mathbf{k}} \cdot \hat{\mathbf{n}}_D > 0$ (forward flux) and $-\mathcal{F}(\mathbf{W}^-)\hat{\mathbf{n}}_D \quad -\mathcal{F}(\mathbf{W}^+)\hat{\mathbf{n}}_D$ if $\hat{\mathbf{k}} \cdot \hat{\mathbf{n}}_D < 0$ (backward flux), where $\mathbf{W}^\pm(\mathbf{x}, \mathbf{k}, t) \quad \lim_{h\downarrow 0} \mathbf{W}(\mathbf{x} \pm h\hat{\mathbf{n}}_D, \mathbf{k}, t)$ for a.e. $\mathbf{x} \in \gamma_D$, and $\mathbf{W}^+(\mathbf{x}, \mathbf{k}, t) \quad \mathbf{0}$ whenever $\mathbf{x} \in \partial\mathcal{D}$; likewise, $c_\alpha^\pm(\mathbf{x}) \quad \lim_{h\downarrow 0} c_\alpha(\mathbf{x} \pm h\hat{\mathbf{n}}_D)$ a.e. $\mathbf{x} \in \gamma_D$. If however the group velocities are discontinuous across γ_D, such that $c_\alpha^+ \;/\; c_\alpha^-$, these expressions may be generalized to the form:

$$\mathcal{F}(\mathbf{W}^+)\hat{\mathbf{n}}_D \quad \mathcal{R}_D^+(\mathbf{W}^+) + \mathcal{T}_D^-(\mathbf{W}^-) \tag{7}$$

for $\mathbf{z} \in \Gamma^+(\gamma_D)$, and:

$$-\mathcal{F}(\mathbf{W}^-)\hat{\mathbf{n}}_D \quad \mathcal{R}_D^-(\mathbf{W}^-) + \mathcal{T}_D^+(\mathbf{W}^+) \tag{8}$$

for $\mathbf{z} \in \Gamma^-(\gamma_D)$, where $\mathcal{R}_D^\pm : \Gamma^\mp(\gamma_D) \to \Gamma^\pm(\gamma_D)$ and $\mathcal{T}_D^\mp : \Gamma^\pm(\gamma_D) \to \Gamma^\pm(\gamma_D)$ are the upstream $(-)$ and downstream $(+)$ reflexion and transmission operators, respectively, of the interface γ_D. These relations show how the inward and outward normal power flows are distributed on the interface: the outward flows are the sum of reflected and transmitted flows, taking into account possible mode conversions in E, on either side of that interface. The explicit derivation of the reflection and transmission operators for junctions of thick beams or shells is outlined in Savin (2007). Their general properties and some examples are also given in Savin (2011).

3. Numerical methods

Numerical methods with low numerical dispersion and dissipation errors are needed to perform long-time simulations of the MRTE in order to reach its diffusion limit. Monte-Carlo methods and discontinuous finite element methods both have these properties. Monte-Carlo methods are very easy to implement since they are based on an intuitive interpretation of the RTEs. They also allow to compute local solutions, a decisive advantage over energetic methods (such as finite elements) for large-scale computations. They converge slowly, but it is always possible to control their accuracy. Their main drawback is their lack of versatility for complex geometries as typically encountered in structural dynamics and acoustics. Finite element methods are, on the contrary, much more flexible and can be applied to truly complex geometries. Among these methods, the discontinuous "Galerkin" (DG) finite element method has originally been introduced by Reed & Hill (1973) and Lesaint & Raviart (1974) to compute neutronic transfers; it is therefore adapted to the integration of RTEs. In the DG method, boundary fluxes at the edges or faces of the elements maintain the consistency of the numerical solution with the continuous, non-discretized RTEs, which are otherwise discretized on each element without any continuity relation between elements. This scheme is recalled in Sect. 3.2 below; the numerical fluxes are constructed in order to get to satisfy the boundary and interface conditions (6) and (7)-(8).

3.1. *Monte-Carlo method*

The Monte-Carlo method for solving a transport partial differential equation or a radiative transfer integro-differential equation is based on its interpretation as a Fokker-Planck equation for the marginal probability density $\boldsymbol{\rho} \quad (\rho_\alpha)_{\alpha\in E}$ of an underlying homogeneous Markov process $\{(\mathbf{Z}_t, A_t),\, t \geq 0\}$ having its values in $T^*\mathbb{R}^d \times E$ (Lapeyre *et al.*, 2003). Let $\mathcal{A}$ be the infinitesimal generator of that process, that is:

$$\mathcal{A}\phi \quad \lim_{h\downarrow 0} \frac{1}{h} \left(\mathbb{E}_{\mathbf{z},\alpha}\{\phi(\mathbf{Z}_h, A_h)\} - \phi(\mathbf{z},\alpha)\right)$$

$\forall\phi \in \mathcal{C}_b^1(T^*\mathbb{R}^d \times E)$, where:

$$\begin{aligned}&\mathbb{E}_{\mathbf{z},\alpha}\{\phi(\mathbf{Z}_t, A_t)\} : \\ &\qquad \mathbb{E}\{\phi(\mathbf{Z}_t, A_t) | (\mathbf{Z}_t, A_t)_{t \quad 0} \quad (\mathbf{z},\alpha)\}\,.\end{aligned}$$

$\mathbb{E}\{\cdot\}$ stands for the mathematical expectation, and $\mathcal{C}_b^1(T^*\mathbb{R}^d \times E)$ is the set of bounded functions which are $\mathcal{C}^1$ with respect to the physical space variable $\mathbf{x} \in \mathbb{R}^d$. The associated Fokker-Planck, or forward Kolmogorov equation is $\partial_t\boldsymbol{\rho}(\mathbf{z},t) \quad \mathcal{A}^*\boldsymbol{\rho}(\mathbf{z},t)$, where $\mathcal{A}^*$ stands for the adjoint operator. This is exactly the MRTE (5) if $\mathcal{A}$ is chosen as $\mathcal{A}\boldsymbol{\rho} \;:\quad (\mathbf{F}_\alpha \cdot \boldsymbol{\nabla}_{\mathbf{z}}\rho_\alpha)_{\alpha\in E} + \mathcal{Q}_{\mathbf{x}}^*(\boldsymbol{\rho})$, where the adjoint $\mathcal{Q}_{\mathbf{x}}^*$ of the collision operator (4) is constructed by taking the transpose matrix $\boldsymbol{\sigma}^{\mathrm{T}}$ as the kernel of $\mathcal{K}_{\mathbf{x}}^*$. Its solution $\boldsymbol{\rho}$ satisfies:

$$\begin{aligned}&\sum_{\alpha\in E}\int_{T^*\mathbb{R}^d} \phi(\mathbf{z},\alpha)\rho_\alpha(\mathrm{d}\mathbf{z},t) \\ &\qquad \sum_{\alpha\in E}\int_{T^*\mathbb{R}^d} \mathbb{E}_{\mathbf{z},\alpha}\{\phi(\mathbf{Z}_t, A_t)\}\rho_\alpha^0(\mathrm{d}\mathbf{z})\end{aligned}$$

$\forall\phi \in \mathcal{C}_b^1(T^*\mathbb{R}^d \times E)$, starting from an initial condition $\boldsymbol{\rho}(\mathbf{z},0) \quad \boldsymbol{\rho}^0(\mathbf{z})$. Then taking formally $\phi(\mathbf{z},\alpha) \quad \delta(\mathbf{z}-\mathbf{z}_0)\otimes\delta(\alpha-\alpha_0)$ (but $\phi \notin \mathcal{C}_b^1(T^*\mathbb{R}^d \times E)$!) and the usual estimator of the mathematical expectation $\mathbb{E}\{\phi(\mathbf{Z}_t, A_t)\} \simeq \frac{1}{N}\sum_{j \quad 1}^N \phi(\mathbf{z}^j(t), \alpha^j(t))$ constructed by simulating N trajectories $t \mapsto (\mathbf{z}^j(t), \alpha^j(t))$, $1 \leq j \leq N$, of the process $\{(\mathbf{Z}_t, A_t),\ t \geq 0\}$, yields:

$$w_{\alpha_0}(\mathbf{z}_0, t) \simeq \frac{\mathcal{E}_0}{N} \sum_{j\quad 1}^{N} \delta(\mathbf{z}^j(t) - \mathbf{z}_0) \otimes \delta(\alpha^j(t) - \alpha_0)\,,$$

with $\mathcal{E}_0 \quad \sum_{\alpha \in E} w_\alpha^0(T^*\mathbb{R}^d)$. Alternatively, if the MRTE (5) is written $\partial_t \mathbf{W} \quad \mathcal{A}\mathbf{W}$ where $\mathcal{A}\mathbf{W} \ : \quad -\boldsymbol{\nabla}_\mathbf{z} \cdot \mathcal{F}(\mathbf{W}) + \mathcal{Q}_\mathbf{x}(\mathbf{W})$, then its solution in $\mathcal{C}_b^1(T^*\mathbb{R}^d)$ is given by Feynman-Kac's formula:

$$\mathbf{W}(\mathbf{z}, t) \quad \mathbb{E}_\mathbf{z}\{\mathbf{W}^0(\mathbf{Z}_t)\}\,. \tag{9}$$

This algorithm extends to a bounded domain $\mathcal{D}$ by constructing the trajectories $t \mapsto (\mathbf{z}^j(t), \alpha^j(t))$ from the broken bicharacteristic flow within $T^*\mathcal{D}$, assuming that its intersection with the glancing set remains empty for all times $t \in \ 0, T$. This is practically done by reflecting and transmitting the energy rays in $\mathcal{D}$ according to Snell-Descartes laws with all possible mode conversions in E.

3.2. *Discontinuous finite element method*

3.2.1. *Basic principles*

Let us now consider the subdivision $\mathcal{T}_h \quad \cup_{r\quad 1}^{\mathcal{N}} \mathcal{D}_r$ of $\mathcal{D}$ into $\mathcal{N}$ non-overlapping sub-domains, or finite elements $\mathcal{D}_r$. A Galerkin variational formulation of the system (5) in an *ad hoc* functional set $\mathcal{W}_h$ defined on $\mathcal{T}_h$ is: Find $\mathbf{W} \in \mathcal{W}_h$ such that for all $\mathbf{v} \in \mathcal{W}_h$

$$\begin{aligned}\int_{\mathcal{D}_r \times \mathbb{R}^d} (\partial_t \mathbf{W} - \mathcal{Q}_\mathbf{x}(\mathbf{W})) \cdot \mathbf{v}\, \mathrm{d}\mathbf{z} \\ - \int_{\mathcal{D}_r \times \mathbb{R}^d} \mathcal{F}(\mathbf{W}) : \boldsymbol{\nabla}_\mathbf{z} \mathbf{v}^\mathrm{T}\, \mathrm{d}\mathbf{z} \\ + \int_{\partial\mathcal{D}_r \times \mathbb{R}^d} \mathcal{F}(\mathbf{W})\hat{\mathbf{n}}_r \cdot \mathbf{v}\, \mathrm{d}\gamma(\mathbf{z}) \quad 0\,,\end{aligned} \tag{10}$$

where $\hat{\mathbf{n}}_r$ is outward unit normal to $\mathcal{D}_r$, and γ is the natural surface measure on $\partial\mathcal{D}_r \times \mathbb{R}^d$. If the set $\mathcal{W}_h$ is continuous over adjacent subdomains, summing the above expression yields the standard continuous variational formulation: For all $\mathbf{v} \in \mathcal{W}_h$

$$\begin{aligned}\int_{T^*\mathcal{D}} (\partial_t \mathbf{W} - \mathcal{Q}_\mathbf{x}(\mathbf{W})) \cdot \mathbf{v}\, \mathrm{d}\mathbf{z} \\ - \int_{T^*\mathcal{D}} \mathcal{F}(\mathbf{W}) : \boldsymbol{\nabla}_\mathbf{z} \mathbf{v}^\mathrm{T}\, \mathrm{d}\mathbf{z} \\ + \int_{\Gamma^+(\partial\mathcal{D})} \mathcal{F}(\mathbf{W})\hat{\mathbf{n}} \cdot \mathbf{v}\, \mathrm{d}\gamma(\mathbf{z}) \\ \int_{\Gamma^-(\partial\mathcal{D})} (\mathbf{f} + \mathcal{R}_\mathbf{x}(\mathbf{W})) \cdot \mathbf{v}\, \mathrm{d}\gamma(\mathbf{z})\,.\end{aligned} \tag{11}$$

If however $\mathcal{W}_h$ is discontinous over two adjacent sub-domains, then the normal flux $\mathcal{F}(\mathbf{W})\hat{\mathbf{n}}_r$ in Eq. (10) has to be replaced by a numerical flux $\mathcal{F}_r^*(\mathbf{W}^-, \mathbf{W}^+)\hat{\mathbf{n}}_r$ because the left and right traces of $\mathbf{W}$ on $\partial\mathcal{D}_r$, denoted by $\mathbf{W}^-$ and $\mathbf{W}^+$ respectively, are different *a priori*. The corresponding discontinuous variational formulation is: For all $\mathbf{v} \in \mathcal{W}_h$

$$\begin{aligned}\int_{\mathcal{D}_r \times \mathbb{R}^d} (\partial_t \mathbf{W} - \mathcal{Q}_\mathbf{x}(\mathbf{W})) \cdot \mathbf{v}\, \mathrm{d}\mathbf{z} \\ - \int_{\mathcal{D}_r \times \mathbb{R}^d} \mathcal{F}(\mathbf{W}) : \boldsymbol{\nabla}_\mathbf{z} \mathbf{v}^\mathrm{T}\, \mathrm{d}\mathbf{z} \\ + \int_{\partial\mathcal{D}_r \times \mathbb{R}^d} \mathcal{F}_r^*(\mathbf{W}^-, \mathbf{W}^+)\hat{\mathbf{n}}_r \cdot \mathbf{v}\, \mathrm{d}\gamma(\mathbf{z}) \quad 0\,,\end{aligned} \tag{12}$$

with $\mathbf{W}^\pm \qquad \lim_{h \downarrow 0} \mathbf{W}(\mathbf{x} \pm h\hat{\mathbf{n}}_r, \mathbf{k}, t)$ a.e. $\mathbf{x} \in \partial\mathcal{D}_r$, and the convention $-\mathcal{F}_r^*(\mathbf{W}^-, \mathbf{W}^+)\hat{\mathbf{n}}_r \quad \mathbf{f} + \mathcal{R}_\mathbf{x}(\mathbf{W}^-)$ and $\mathbf{W}^+ \quad \mathbf{0}$ if $\mathbf{z} \in \Gamma^-(\partial\mathcal{D})$. The numerical flux has also to be consistent, that is $\mathcal{F}_r^*(\mathbf{W}, \mathbf{W}) \quad \mathcal{F}(\mathbf{W})$, and conservative, that is $\mathcal{F}_r^*(\mathbf{W}^-, \mathbf{W}^+)\hat{\mathbf{n}}_r + \mathcal{F}_s^*(\mathbf{W}^-, \mathbf{W}^+)\hat{\mathbf{n}}_s \quad \mathbf{0}$ if $\partial\mathcal{D}_r \cap \partial\mathcal{D}_s \ / \ 0$. The above formulation is the so-called "weak" form of the DG method (Hesthaven & Warburton, 2008). The "strong" form corresponding to the original DG method introduced by Lesaint & Raviart (1974) is derived by integrating the weak form by parts. Let $\mathcal{L}\mathbf{W} \ :$

$\partial_t \mathbf{W} + \boldsymbol{\nabla}_{\mathbf{z}} \cdot \mathcal{F}(\mathbf{W})$, then: For all $\mathbf{v} \in \mathcal{W}_h$

$$\int_{\mathcal{D}_r \times \mathbb{R}^d} (\mathcal{L}\mathbf{W} - \mathcal{Q}_{\mathbf{x}}(\mathbf{W})) \cdot \mathbf{v}\, \mathrm{d}\mathbf{z} \quad \int_{\partial\mathcal{D}_r \times \mathbb{R}^d} \left(\mathcal{F}(\mathbf{W}^-) - \mathcal{F}_r^*(\mathbf{W}^-, \mathbf{W}^+)\right) \hat{\mathbf{n}}_r \cdot \mathbf{v}\, \mathrm{d}\gamma\,. \tag{13}$$

This last form can also be seen as a penalization method applied to the boundary fluxes of the sub-domains (or elements). An alternative form for linear systems is the ultra-weak variational formulation proposed by Després (1994). It consists in choosing $\mathcal{W}_h \quad \{\mathbf{v}\,,\ \mathcal{L}^*\mathbf{v} \quad \mathbf{0}\}$, where $\mathcal{L}^*\mathbf{W} \equiv -\partial_t \mathbf{W} - (\mathbf{F}_\alpha \cdot \boldsymbol{\nabla}_{\mathbf{z}} w_\alpha)_{\alpha \in E}$ is the formal adjoint of $\mathcal{L}$. Then (13) is written: For all $\mathbf{v} \in \mathcal{W}_h$

$$\partial_t \left(\int_{\mathcal{D}_r \times \mathbb{R}^d} \mathbf{W} \cdot \mathbf{v}\, \mathrm{d}\mathbf{z} \right) + \int_{\partial\mathcal{D}_r \times \mathbb{R}^d} \mathcal{F}_r^*(\mathbf{W}^-, \mathbf{W}^+) \hat{\mathbf{n}}_r \cdot \mathbf{v}\, \mathrm{d}\gamma(\mathbf{z}) \quad \int_{\mathcal{D}_r \times \mathbb{R}^d} \mathcal{Q}_{\mathbf{x}}(\mathbf{W}) \cdot \mathbf{v}\, \mathrm{d}\mathbf{z}\,. \tag{14}$$

The test functions $\mathbf{v}$ have then to be taken as solutions of an adjoint problem, typically plane waves in piecewise homogeneous media (Gabard, 2007): $\mathbf{v}(\mathbf{z}, t) \quad \mathbf{V}\phi(\mathbf{k} \cdot \mathbf{x} - \omega t)$ such that $\boldsymbol{\Gamma}(\mathbf{k})\mathbf{V} \quad \omega\mathbf{V}$, meaning that ω and $\mathbf{V}$ are the eigenvalues and eigenvectors of the acoustic tensor $\boldsymbol{\Gamma}$ which is simply $\boldsymbol{\Gamma}(\mathbf{k}) \quad \mathrm{diag}(\lambda_\alpha(\mathbf{x}, \mathbf{k}))_{\alpha \in E}$ here.

3.2.2. *Numerical implementation*

The numerical implementation of the DG method expounded above requires three main ingredients, once the subdivision $\mathcal{T}_h$ has been performed: (i) the definition of numerical fluxes $\mathcal{F}_r^*(\mathbf{W}^-, \mathbf{W}^+)$ on all elements, (ii) the choice of the approximation set $\mathcal{W}_h$, and (iii) the choice of a time-integration scheme. The computational domain in phase-space is reduced to the unit cosphere bundle $S^*\mathcal{D}$ $\mathcal{D} \times \mathbb{S}^{d-1}$ by considering only normalized wave vectors $\hat{\mathbf{k}}$ defined on the unit sphere $\mathbb{S}^{d-1}$ of $\mathbb{R}^d$. For applications to assemblies of slender structures, $d \quad 1$ (beams) and $\mathbb{S}^0 \quad \{-1, +1\}$, or $d \quad 2$ (shells) and the unit circle $\mathbb{S}^1$ is parametrized by $\theta \in \ 0, 2\pi\ $ such that $\hat{\mathbf{k}} \quad (\cos\theta, \sin\theta)^{\mathrm{T}}$. In both cases the meshing of $S^*\mathcal{D}$ writes $\mathcal{T}_h \quad \cup_{r\ 1}^{\mathcal{N}} X_r \quad \cup_{r\ 1}^{\mathcal{N}} \mathcal{D}_r \times \mathbb{S}^{d-1}$, denoting by h the largest diameter of the elements $\mathcal{D}_r$.

Regarding (i), the numerical fluxes ensuring the stability and consistency of the DG method for transport equations have the form (Brezzi *et al.*, 2004):

$$\mathcal{F}_r^*(\mathbf{W}^-, \mathbf{W}^+) \quad \frac{1}{2}\left(\mathcal{F}(\mathbf{W}^-) + \mathcal{F}(\mathbf{W}^+)\right) + \mathbf{C}_r \left(\mathbf{W}^- - \mathbf{W}^+\right) \hat{\mathbf{n}}_r$$

where $\mathbf{C}_r$ is an $M \times M$ non-negative definite matrix. For example, the (upwind) Roe flux corresponds to $\mathbf{C}_r \quad \frac{1}{2}\mathrm{diag}(c_\alpha |\hat{\mathbf{k}} \cdot \hat{\mathbf{n}}_r|)_{\alpha \in E}$ and the Lax-Friedrichs flux corresponds to $\mathbf{C}_r \quad \frac{1}{2}\mathrm{diag}(c_\alpha)_{\alpha \in E}$. This expression can be generalized as follows for an element boundary $\partial\mathcal{D}_r$ belonging to an interface (a junction) γ_D where the group velocities c_α are discontinuous. Let $\gamma_{Dr} \quad \partial\mathcal{D}_r \cap \gamma_D \ / \ \emptyset$, then having in mind Eq. (7) and Eq. (8), the numerical flux is written on $\Gamma^+(\gamma_{Dr})$:

$$\mathcal{F}_r^*(\mathbf{W}^-, \mathbf{W}^+)\hat{\mathbf{n}}_r \quad \left(\frac{1}{2} - C_r\right) \mathcal{F}(\mathbf{W}^-)\hat{\mathbf{n}}_r + \left(\frac{1}{2} + C_r\right) \left(\mathcal{R}_D^+(\mathbf{W}^+) + \mathcal{T}_D^-(\mathbf{W}^-)\right) \tag{15}$$

and on $\Gamma^-(\gamma_{Dr})$:

$$\mathcal{F}_r^*(\mathbf{W}^-, \mathbf{W}^+)\hat{\mathbf{n}}_r \quad \left(\frac{1}{2} - C_r\right) \mathcal{F}(\mathbf{W}^+)\hat{\mathbf{n}}_r - \left(\frac{1}{2} + C_r\right) \left(\mathcal{R}_D^-(\mathbf{W}^-) + \mathcal{T}_D^+(\mathbf{W}^+)\right)\,, \tag{16}$$

with $0 \leq |C_r| \leq \frac{1}{2}$. This definition extends in fact to the case $\partial\mathcal{D}_r \cap \gamma_D \quad \emptyset$

for which $\mathcal{R}_D^{\pm}(\mathbf{W}^{\pm}) \quad 0$ and $\mathcal{T}_D^{\pm}(\mathbf{W}^{\pm}) \quad \mp\mathcal{F}(\mathbf{W}^{\mp})\hat{\mathbf{n}}_D$ on $\Gamma^{\mp}(\partial\mathcal{D}_r)$:

$$\mathcal{F}_r^*(\mathbf{W}^-,\mathbf{W}^+) \quad \left(\frac{1}{2}-C_r\right)\mathcal{F}(\mathbf{W}^-) + \left(\frac{1}{2}+C_r\right)\mathcal{F}(\mathbf{W}^+)$$

for $\mathbf{z} \in \Gamma^+(\partial\mathcal{D}_r)$, and:

$$\mathcal{F}_r^*(\mathbf{W}^-,\mathbf{W}^+) \quad \left(\frac{1}{2}-C_r\right)\mathcal{F}(\mathbf{W}^+) + \left(\frac{1}{2}+C_r\right)\mathcal{F}(\mathbf{W}^-)$$

for $\mathbf{z} \in \Gamma^-(\partial\mathcal{D}_r)$.

Regarding (ii), the approximation set $\mathcal{W}_h \subset L^2(S^*\mathcal{D})$ is constructed by tensoring the physical space and phase variables $\mathcal{W}_h \quad \mathcal{D}_h \otimes \mathcal{S}_h$, with:

$$\mathcal{D}_h \quad \{\mathbf{v} \in (L^2(\mathcal{D})^M;\, v_\alpha|_{\mathcal{D}_r} \in \mathbb{P}_p(\mathcal{D}_r),\ 1 \leq r \leq \mathcal{N},\, \alpha \in E\}$$

where $\mathbb{P}_p(\mathcal{D}_r) \subset H^1(\mathcal{D}_r)$ is typically a polynomial set, and $\mathcal{S}_h$ is a finite dimensional set of real-valued functions defined on $\mathbb{S}^{d-1}$. Practically in dimension 1, $\mathbb{P}_p(\mathcal{D}_r)$ is spanned by either a set of Jacobi orthogonal polynomials up to the order p (the modal, or spectral approach), or a set of Lagrange interpolation polynomials associated to $p+1$ interpolation points (the nodal approach). The degrees of freedom in physical space are the projection coefficients on the Jacobi polynomials of the approximate solution in $\mathcal{W}_h$ in the modal approach, and its values at the interpolation points in the nodal approach. It should be noted that if the latter are chosen as Gauss-Jacobi quadrature points on $-1,1$, the corresponding Lagrange polynomials are computed explicitly as functions of Jacobi polynomials, which yields a unified framework for the modal and nodal approaches. The numerical dispersion and dissipation errors introduced by this spatial discretization are $\mathrm{O}(K^{2p+3})$ and $\mathrm{O}(K^{2p+2})$, respectively, for decentered numerical fluxes ($C_r \;/\; 0$), or $\mathrm{O}(K^{4\,\frac{p}{2}\,+3})$ and 0 for centered fluxes ($C_r \quad 0$) (Ainsworth, 2004), where $K \quad h|\mathbf{k}|$ and $\cdot$ stands for the integer part. These so-called "superconvergence" properties of DG methods are particularly attractive for long-times numerical simulations. The spatial discretization in dimension 2 is achieved by tensoring both dimensions on structured meshes, as proposed in Cohen *et al*. (2006) for example. The construction of $\mathcal{S}_h$ is however more involved. The modal approach corresponds to a truncated Fourier expansion in θ, but this scheme does not ensure the positivity of the approximate solution because of the slowness of the convergence of Fourier series. The nodal approach by Lagrange interpolation yields the well-known ray effect which also arises in the discrete ordinate methods widely used in neutronic or thermal transfer computations (Duderstadt & Martin, 1979, chap. 8). Therefore further researches are needed on this issue; it is the subject of ongoing investigations.

At last, regarding (iii), strong stability-preserving (SSP), high-order Runge-Kutta schemes have been implemented (Gottlieb *et al*., 2001). The CFL condition of these schemes, coupled to the above mentioned phase-space discretization methods, can be computed numerically for various orders p.

4. Numerical examples

In the examples below the radiative transfer equations (1) with specular-like boundary and interface (at the junctions) conditions are solved numerically by the schemes expounded in the foregoing section. In Sect. 4.1 the example of Savin (2007) of scalar transport in coupled Timoshenko beams is resumed and extended to randomly heterogeneous beams. Sect. 4.2 deals with two coupled thick cylindrical shells.

4.1. *Coupled beams*

We first consider an assembly of two random Timoshenko beams $\quad -L, 0 \cup 0, L$, $L > 0$, with a junction at $x \quad 0$ such that beam #2 forms an angle ϕ with beam #1. Three types of energy rays propagate in such a system: a pure shear transverse mode at the velocities $c_{\mathrm{T}j}$ ($j \quad 1$ or 2 for either beam #1 or beam #2) denoted by subscript T, a pure bending mode at the velocities $c_{\mathrm{P}j}$ denoted by subscript Pb, and a longitudinal mode at the velocities $c_{\mathrm{P}j}$ denoted by subscript Pn; see Savin (2005b, 2008) for some details on the derivation. Therefore the energy density evolution in this system is depicted by three coupled, one-dimensional radiative transfer equations ($M \quad 3$ and $d \quad 1$ such that $\hat{k} \equiv \mathrm{sign}(k)$) for $\alpha \quad \mathrm{T}, \mathrm{Pb}$ or Pn written as:

$$\partial_t w_\alpha + c_\alpha \hat{k} \partial_x w_\alpha + \eta \Sigma_\alpha w_\alpha \qquad \Sigma_\alpha(k)\left(w_\alpha(-k) - w_\alpha(k)\right) , \quad (17)$$

where the coupling is ensured by the left ($-$) and right ($+$) power reflection and transmission operators $\mathcal{R}_D^\pm$, $\mathcal{T}_D^\pm$ of the junction. Indeed, for thick beams no coupling of modes occurs through the collision operator on the right-hand-side in Eq. (17) (Savin, 2008). The expressions of the kernels of $\mathcal{R}_D^\pm$ and $\mathcal{T}_D^\pm$ as functions of ϕ and beams mechanical and geometrical characteristics are derived in Savin (2007), using a high-frequency model compatible with the propagation characteristics obtained by the transport theory. This study has also shown that the power reflection coefficients at a beam end for either Dirichlet (clamped) or Neumann (free) boundary condition were all equal to 1, without mode conversion of any type; therefore specular reflections are considered at the boundaries $x \quad \pm L$. In agreement with their analytical expressions derived in Savin (2005b, 2008), the total scattering cross-sections are written $\Sigma_\alpha(k) \quad \frac{\pi}{2} c_\alpha k^2 \rho_\alpha S(2k)$, where the phase function $S(k)$ is the one-dimensional Fourier transform of the normalized correlation function $R(x-y)$ of random fluctuations of the density and inverse Young's and shear modulii, $S(k) \quad \frac{1}{2\pi}\int_{\mathbb{R}} R(r) e^{ikr} \mathrm{d}r$; and ρ_α is a dimensionless strength factor related to their variation coefficients (or amplitudes). Its analytical expressions for various correlation models are gathered in Table 1 below, where the correlation function $R(r)$ is normalized such that $a \; : \quad \int_0^{+\infty} R(r)\mathrm{d}r$ is a correlation length of these random fluctuations.

All computations are performed for the initial conditions:

$$w_{\mathrm{T}}^0(x,k) \quad \varphi\left(\frac{x-x_0}{r_0}\right) \otimes \delta(\hat{k}-\hat{k}_0) ,$$
$$w_{\mathrm{Pb}}^0(x,k) \quad w_{\mathrm{Pn}}^0(x,k) \equiv 0 ,$$

with $\varphi(x) \quad \max(0, 1-2|x|)$ ("hat" function). The parameters are chosen as $x_0 \quad -0.1 \times L$, $r_0 \quad 0.12 \times L$, and $\hat{k}_0 \quad +1$. Beam velocities are such that $c_{\mathrm{T}2}/c_{\mathrm{T}1} \quad c_{\mathrm{P}2}/c_{\mathrm{P}1} \quad \sqrt{2}$, their Poisson's ratio being $\nu_1 \quad \nu_2 \quad 0.3$, and the junction angle is $\phi \quad \pi/3$.

The coupled radiative transfer equations (17) are solved numerically by the Runge-Kutta DG finite element method expounded in Sect. 3.2 above. The assembly is partitioned into 40 uniform spatial elements. Legendre polynomials up to the fifth order are used as local basis functions on them, and Eqs. (17) are solved in time by a fourth-order SSP Runge-Kutta scheme. The Courant number for the simulations is fixed at $\mathrm{c} \equiv c_{\mathrm{T}2}\frac{\Delta t}{\Delta x} \quad 2.5\,10^{-2}$. Fig. 1 through Fig. 3 display our results for $\rho_\alpha \quad 0.3$, $ak \quad 2$, and Gaussian, Markov and Rayleigh models of correlation for the scattering cross-sections, respectively. The first case corresponds to smooth,

Table 1. Phase function models for beams; K_v is the modified Bessel function of the second kind, Γ is the Gamma function, $v \in \;\; 0,1\;\;$ is the Hurst number in von Karman model (note that the latter reduces to Markov model for $v \quad \frac{1}{2}$).

Correlation model	$R(r)$	$S(k)$
Gaussian	$\exp(-\pi\frac{r^2}{4a^2})$	$\frac{a}{\pi}\exp(-\frac{a^2k^2}{\pi})$
Markov	$\exp(-\frac{\lvert r\rvert}{a})$	$\frac{a}{\pi}(1+a^2k^2)^{-1}$
von Karman	$\frac{2^{1-v}}{\sqrt{\pi}\,\Gamma(v+\frac{1}{2})}\left(\frac{\lvert r\rvert}{a}\right)^v K_v(\frac{\lvert r\rvert}{a})$	$\frac{a}{\pi}(1+a^2k^2)^{-(v+\frac{1}{2})}$
Rayleigh	$\delta(\frac{\lvert r\rvert}{2a})$	$\frac{a}{\pi}$

differentiable random heterogeneities (in the mean-square sense), while the second and third ones correspond to non-differentiable heterogeneities. The time scale is $T \quad L/c_{\mathrm{T2}}$, the time needed by the wave front to reach the extremity $x \quad L$ of the homogeneous assembly starting from the junction. The thin vertical line indicates the location of the junction. The wavefronts of shear ($\alpha \quad$ T) and bending/axial ($\alpha \quad$ P) energy densities propagating at the speeds $c_{\mathrm{T}j}$ and $c_{\mathrm{P}j}$, respectively, are marked by small capital letters. A small amount of damping $\eta \quad 0.01$ has been added such that the total energy within the system at a fixed wavenumber k evolves as (see Eq. (3)):

$$\mathcal{E}(t) \qquad \sum_{\alpha \quad \mathrm{T,Pb,Pn}} e^{-\eta\Sigma_\alpha t}\mathcal{E}_\alpha^0\,, \qquad \mathcal{E}_\alpha^0 \qquad \int_{-L}^{L} w_\alpha^0(x,\pm|k|)\mathrm{d}x\,. \tag{18}$$

This exact result is recovered numerically by our scheme, as shown by Fig. 4 through Fig. 6. They display the overall mechanical energies $\mathcal{E}_j^h(t)$ computed within each beam by $\mathcal{E}_1^h(t) \quad \int_{-L}^{0}\mathcal{E}^h(x,t)\mathrm{d}x$, $\mathcal{E}_2^h(t) \quad \int_0^L \mathcal{E}^h(x,t)\mathrm{d}x$, and their sum, normalized by the total energy introduced into the system $\mathcal{E}(0) \quad \sum_\alpha \mathcal{E}_\alpha^0 \quad \mathcal{E}_{\mathrm{T}}^0$. The computed and exact total energies perfectly match on these plots: one concludes that the proposed scheme in dimension one is numerically conservative.

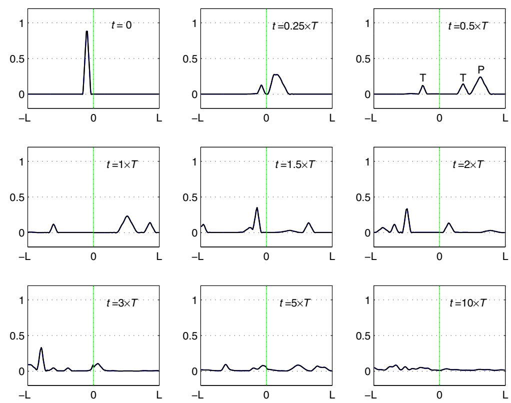

Fig. 1. Evolution of the energy density in an assembly of two randomly heterogeneous Timoshenko beams with $\phi \quad \frac{\pi}{3}$, $c_{\mathrm{T2}}/c_{\mathrm{T1}} \quad c_{\mathrm{P2}}/c_{\mathrm{P1}} \quad \sqrt{2}$, $\nu_1 \quad \nu_2 \quad 0.3$, $\eta \quad 0.01$, $T \quad L/c_{\mathrm{T2}}$, and $\rho_\alpha \quad 0.3$, $ak \quad 2$ for the scattering cross-sections with a Gaussian model of correlation.

These results exhibit the following features: one observes a beating phenomenon of the total energy within each subsystem, by which it is transferred back and forth from the first beam to the second one and the other way round. At late time an equipartition regime holds between both beams. It is described by diffusion equations as explained in Savin (2008) for example. It should be noted that this diffusive regime also arises in the case of a piecewise homogeneous assembly, the onset of diffusion being prompted by the mixing of wave fronts and modes at the junc-

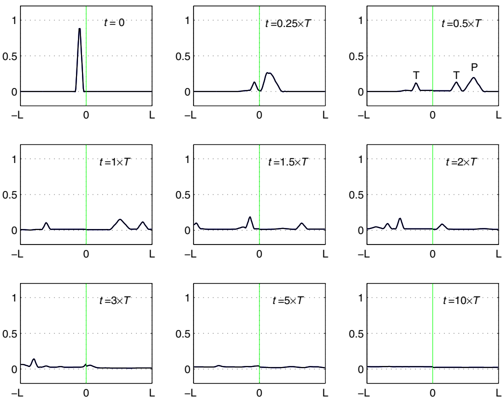

Fig. 2. Same as Fig. 1, with a Markov model of correlation.

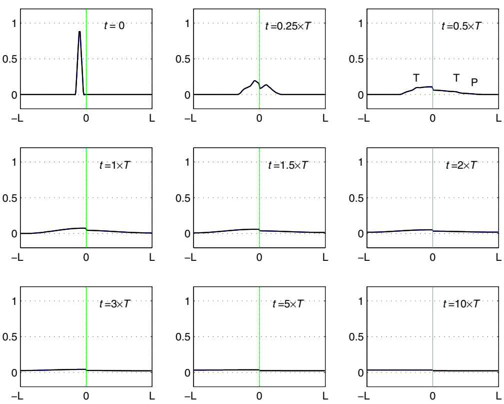

Fig. 3. Same as Fig. 1, with a Rayleigh model of correlation.

tion, and to a lesser extent at the boundaries. This mixing process is only accelerated by the presence of random heterogeneities, as outlined elsewhere (Savin, 2011); also the rougher they are, the faster diffusion arises, and backscattering increases. Finally, the long-time limit of the ratio $\mathcal{E}^h_{\mathrm{P}j}(t)/\mathcal{E}^h_{\mathrm{T}j}(t)$ of the bending/longitudinal energy over the shear energy within each beam is proved to be the ratio $c_{\mathrm{T}j}/c_{\mathrm{P}j}$ to the power of the physical dimension, d 1 in the present case, independently of the source type and scatterers (Papanicolaou & Ryzhik, 1999, Savin, 2008). Here this universal equilibration rule is recovered by the DG

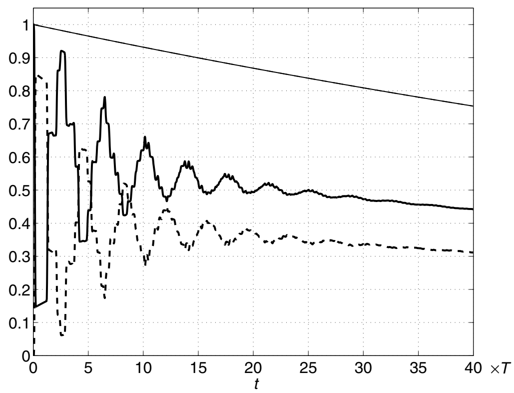

Fig. 4. Evolution of the total vibrational energies in an assembly of two randomly heterogeneous Timoshenko beams with the same data as in Fig. 1, and a Gaussian model of correlation. Thick solid line: beam #1, thick dashed line: beam #2, thin solid line: computed total energy, thin dashed line: exact total energy.

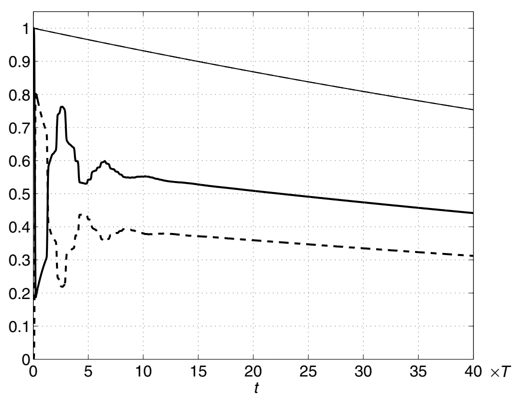

Fig. 5. Same as Fig. 4, with a Markov model of correlation.

scheme, as exemplified by Fig. 7 for a Gaussian model of correlation. The other correlation models yield the same conclusions (plots not displayed here).

4.2. *Coupled shells*

Next we consider an assembly of two thick cylindrical shells, the first one with a constant radius (shell #1) and the second one with a variable radius (shell #2). Both cylinders have the same length L

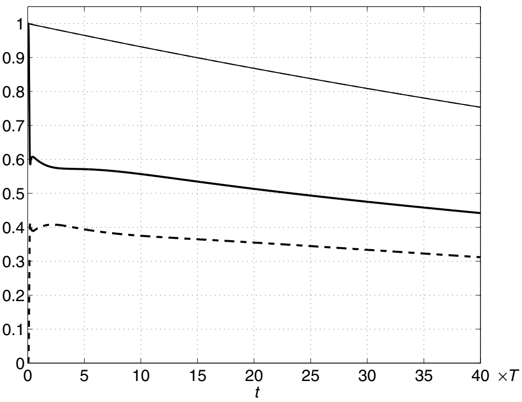

Fig. 6. Same as Fig. 4, with a Rayleigh model of correlation.

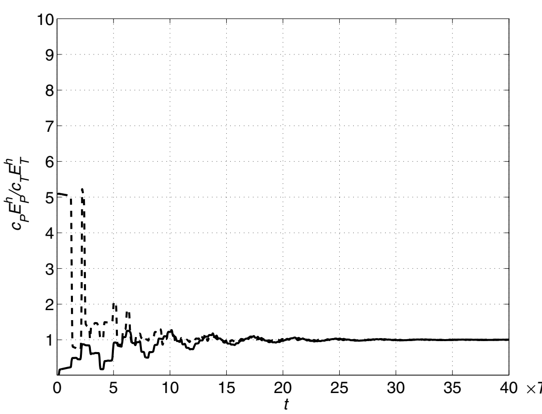

Fig. 7. Evolution of the ratio $c_{\mathrm{P}j}\mathcal{E}^h_{\mathrm{P}j}(t)/c_{\mathrm{T}j}\mathcal{E}^h_{\mathrm{T}j}(t)$ of the total bending/longitudinal and transverse energies in an assembly of two randomly heterogeneous Timoshenko beams with the same data as in Fig. 1, and a Gaussian model of correlation. Solid line: beam #1 ($j = 1$), dashed line: beam #2 ($j = 2$).

which is also twice the constant radius of the first shell. The angle between the shells is again denoted by ϕ, and their mean surfaces are denoted by $\mathcal{D}_1$ and $\mathcal{D}_2$, respectively. Five types of energy rays propagate in such a system: a pure shear transverse mode at the velocities $c_{\mathrm{T}j}$ ($j = 1$ or 2 for either shell #1 or shell #2) denoted by subscript T, and coupled longitudinal and transverse modes P and S each of multiplicity 2 corresponding to either in-plane (subscript n) or bending energies (subscript b), at the velocities $c_{\mathrm{P}j}$ and $c_{\mathrm{S}j}$ respectively; see Savin (2004) for some details on the derivation. Then the energy density evolution in this system is depicted by five coupled radiative transfer equations (1) with $M = 5$–since $\alpha = \mathrm{T}, \mathrm{Pb}, \mathrm{Pn}, \mathrm{Sb}$ or Sn–and $d = 2$, where the coupling is ensured by the left ($-$) and right ($+$) power reflection and transmission operators $\mathcal{R}^{\pm}_D$, $\mathcal{T}^{\pm}_D$ of the shell junction. Their expressions as functions of ϕ, the incident tangent wave vector $\mathbf{k}_t = (\mathbf{I}_2 - \hat{\mathbf{n}}(\mathbf{x}_t) \otimes \hat{\mathbf{n}}(\mathbf{x}_t))\mathbf{k}$ on the junction line $\gamma_D \ni \mathbf{x}_t$, and shells mechanical and geometrical characteristics are derived in Savin (2007), using a high-frequency model compatible with the propagation characteristics obtained by the transport theory within plates. As the curvature tensor has no influence on the high-frequency regime in this model, the same reflection/transmission kernels as for flat plates are used in this example. Homogeneous Neumann boundary conditions are considered at the extremities of the shells, and the corresponding power reflection coefficients have been derived in Savin (2007) as well. Here it has been shown that the shear mode is uncoupled to the in-plane and bending modes by boundary reflections, however the longitudinal P and transverse S modes are coupled. Otherwise all these modes are fully coupled by the junction. The phase function $S(\mathbf{k})$ for the scattering cross-sections derived in Savin (2004, 2008) for shells is now the two-dimensional Fourier transform of the normalized correlation function $R(\mathbf{x} - \mathbf{y})$ of random fluctuations of the density and inverse Lamé's modulii, $S(\mathbf{k}) = \frac{1}{4\pi^2}\int_{\mathbb{R}^2} R(\mathbf{r})e^{\mathrm{i}\mathbf{k}\cdot\mathbf{r}}\mathrm{d}\mathbf{r}$. Analytical expressions of the phase function for various correlation models are gathered in Table 2 below, where the correlation function $R(\mathbf{r})$ is normalized such that $\pi a^2 := \int_0^{+\infty}\int_{\mathbb{S}^1} R(r\hat{\mathbf{r}})r\mathrm{d}r\mathrm{d}\mu(\hat{\mathbf{r}})$.

Table 2. Phase function models for shells (note that von Karman model reduces to Markov model for the Hurst number $\upsilon = \frac{1}{2}$).

Correlation model	$R(\mathbf{r})$	$S(\mathbf{k})$
Gaussian	$\exp(-\frac{\lvert\mathbf{r}\rvert^2}{a^2})$	$\frac{a^2}{4\pi}\exp(-\frac{a^2\lvert\mathbf{k}\rvert^2}{4})$
Markov	$\exp(-\sqrt{2}\frac{\lvert\mathbf{r}\rvert}{a})$	$\frac{a^2}{4\pi}(1+\frac{1}{2}a^2\lvert\mathbf{k}\rvert^2)^{-\frac{3}{2}}$
von Karman	$\frac{2^{-\upsilon}}{\Gamma(\upsilon+1)}\left(\sqrt{2}\frac{\lvert\mathbf{r}\rvert}{a}\right)^{\upsilon}K_\upsilon(\sqrt{2}\frac{\lvert\mathbf{r}\rvert}{a})$	$\frac{a^2}{4\pi}(1+\frac{1}{2}a^2\lvert\mathbf{k}\rvert^2)^{-\upsilon-1}$
Rayleigh	$\frac{\pi}{2}\delta(\frac{\lvert\mathbf{r}\rvert}{a})$	$\frac{a^2}{4\pi}$

The total scattering cross-sections (2) have the form (Savin, 2004, 2008):

$$\Sigma_\alpha(\mathbf{k}) = \frac{\pi}{2}\rho_\alpha c_\alpha|\mathbf{k}|^3\sum_\beta\int_{\mathbb{S}^1} f_{\alpha\beta}(\hat{\mathbf{k}}\cdot\hat{\mathbf{k}}')\times S\left(|\mathbf{k}|\left(\hat{\mathbf{k}}-\frac{c_\alpha}{c_\beta}\hat{\mathbf{k}}'\right)\right)\mathrm{d}\mu(\hat{\mathbf{k}}')\,,$$

where $f_{\alpha\beta}$ is a polynomial function.The initial condition is an isotropic shear impact $w_{\mathrm{T}}^0(\mathbf{x},\mathbf{k}) = \delta(\mathbf{x}-\mathbf{x}_0)$, independently of $\mathbf{k}$, all other modes being initially unloaded. Here $\mathbf{x}_0 = (-L/2, 0)$ and the junction line is the circle $\gamma_D = \{\mathbf{x} = (0,y), 0 \leq y \leq \pi L\}$ in curvilinear coordinates. The Young's modulii, densities, and thicknesses of both shells are equal, such that (Savin, 2004):

$$\frac{c_{\mathrm{P}2}}{c_{\mathrm{P}1}} = \sqrt{\frac{1-\nu_1^2}{1-\nu_2^2}}\,,\quad \frac{c_{\mathrm{S}2}}{c_{\mathrm{S}1}} = \sqrt{\frac{1-\nu_1}{1-\nu_2}}\,,\quad \frac{c_{\mathrm{T}2}}{c_{\mathrm{T}1}} = \sqrt{\frac{6+5\nu_1}{6+5\nu_2}}\,;$$

their Poisson's ratios are $\nu_1 = 0.3$ and $\nu_2 = 0.2$, and the junction angle is $\phi = \pi/12$.

The coupled radiative transfer equations (1) are solved in this case by a direct Monte-Carlo method (Lapeyre *et al.*, 2003). The computer code developed for this application has been fully vectorized, in such a way that numerical simulations of $\mathrm{O}(10^6)$ ray paths with $\mathrm{O}(10^2)$ time steps can be performed in less than one hour of CPU time on a personal computer. Fig. 8 through Fig. 12 below display our results for $a|\mathbf{k}| = 2$, a perturbation strengh $\rho_{\alpha 1} = 0.3$ with a Gaussian or Rayleigh model of correlation in shell #1, and no perturbation $\rho_{\alpha 2} = 0$ in shell #2. The time scale is $T = L/c_{\mathrm{T}2}$, and a small amount of damping $\eta = 0.01$ has been added as for the beam assembly of the previous example. Fig. 8 and Fig. 9 display the evolution of the energy density within the shell assembly for the Gaussian and Rayleigh models of correlation, respectively, while Fig. 10 and Fig. 11 display the evolution of the overall mechanical energies within each shell and their sum.

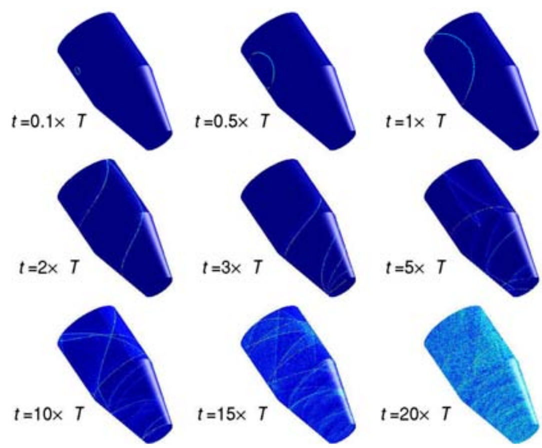

Fig. 8. Evolution of the energy density in an assembly of two randomly heterogeneous cylindrical shells with $\phi = \frac{\pi}{12}$, $\nu_1 = 0.3$, $\nu_2 = 0.2$, $\eta = 0.01$, $T = L/c_{\mathrm{T}2}$, and $\rho_\alpha = 0.3$, $a|\mathbf{k}| = 2$ for the scattering cross-sections with a Gaussian model of correlation in shell #1.

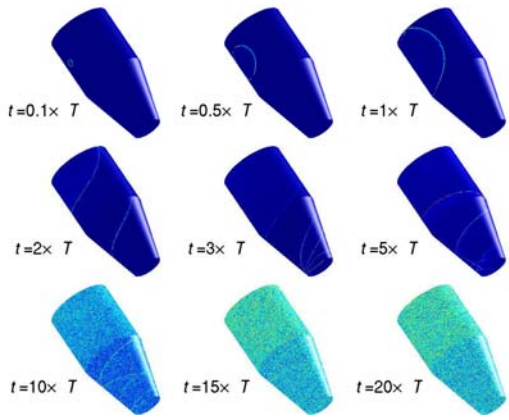

Fig. 9. Same as Fig. 8, with a Rayleigh model of correlation in shell #1.

They show that the Monte-Carlo scheme is numerically conservative. In addition Fig. 12 displays the ratios $c^2_{\mathrm{P}j}\mathcal{E}^h_{\mathrm{P}j}(t)/c^2_{\mathrm{S}j}\mathcal{E}^h_{\mathrm{S}j}(t)$ of the total bending/longitudinal and transverse energies within each shell, respectively, where:

$$\mathcal{E}^h_{\mathrm{P}j}(t) \quad \sum_{\alpha \quad \mathrm{Pn,Pb}} \int_{\mathcal{D}_j\times\mathbb{S}^1} w^h_\alpha(\mathbf{x},\mathbf{k},t)\mathrm{d}\mathbf{x}\,\mathrm{d}\mu(\hat{\mathbf{k}})\,,$$

j 1, 2, and a similar expression for $\mathcal{E}^h_{\mathrm{S}j}(t)$.

All computations exhibit the same features as for the assembly of beams, calling for similar qualitative comments. For an isotropic collision operator, the diffusive limit at large times of $\mathcal{E}^h_{\mathrm{P}j}(t)/\mathcal{E}^h_{\mathrm{S}j}(t)$ is proved again to be the ratio $c_{\mathrm{S}j}/c_{\mathrm{P}j}$ to the power of the physical dimension d 2 independently of the source type and scatterers (Papanicolaou & Ryzhik, 1999, Savin, 2008). This universal equilibration rule is recovered by the Monte-Carlo scheme as well, see Fig. 12. The Markov and von Karman correlation models yield the same quantitative and qualitative conclusions (plots not displayed here). It should be reminded, however, that these results ignore the contribution of the glancing rays possibly trapped by the boundaries and the junction line, because a theoretical model lacks at present to describe them. Indeed it is known from experiments that the vibrational energy is very much likely to be guided by the interfaces, stiffeners or junctions in a real-life complex structure, so these effects should not be disregarded. Further developments are needed on this issue.

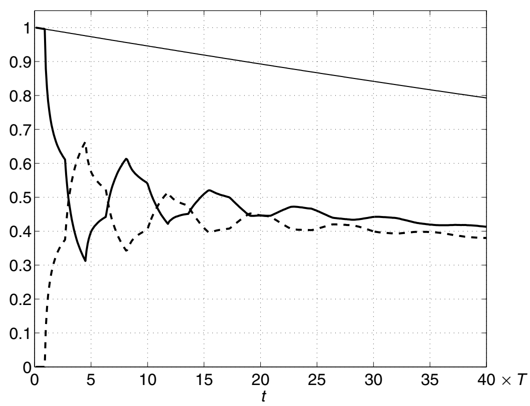

Fig. 10. Evolution of the total vibrational energies in an assembly of two randomly heterogeneous cylindrical shells with the same data as in Fig. 8, and a Gaussian model of correlation in shell #1. Thick solid line: shell #1, thick dashed line: shell #2, thin solid line: computed total energy.

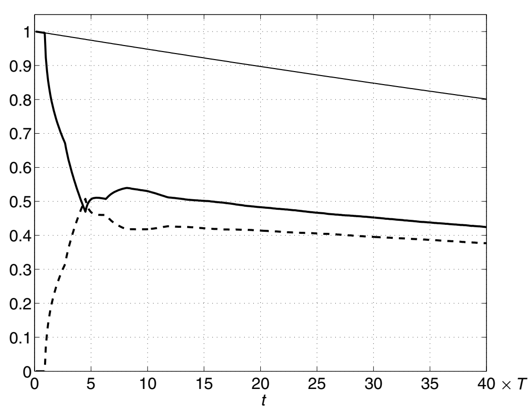

Fig. 11. Same as Fig. 10, with a Rayleigh model of correlation in shell #1.

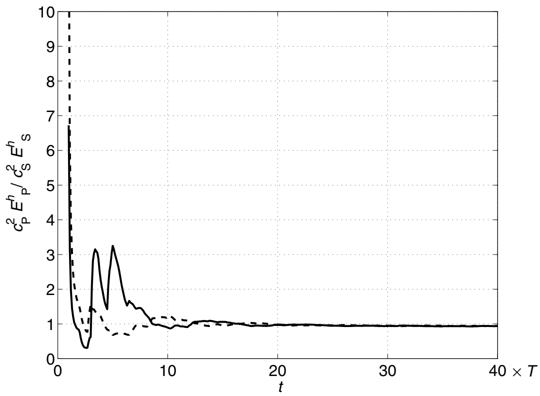

Fig. 12. Evolution of the ratio $c^2_{\mathrm{P}j}\mathcal{E}^h_{\mathrm{P}j}(t)/c^2_{\mathrm{S}j}\mathcal{E}^h_{\mathrm{S}j}(t)$ of the total bending/ longitudinal and transverse energies in an assembly of two randomly heterogeneous cylindrical shells with the same data as in Fig. 8, and a Gaussian model of correlation in shell #1. Solid line: shell #1 (j 1), dashed line: shell #2 (j 2).

5. Conclusions

A radiative transfer model for the vibrational energy evolution in randomly heterogeneous slender structures has been proposed. The radiative transfer model equations have been solved numerically by a discontinuous finite element method and a Monte-Carlo method, among other possible schemes. Both methods have been implemented and tested on examples of assemblies of random thick beams or cylindrical shells. Direct numerical simulations illustrate the transfer regime for these systems, and the onset of a diffusive regime at late times. The latter is characterized by energy equilibration rules very much comparable to the assumption of modal equipartition invoked in the statistical energy analysis of structural-acoustics systems (Lyon & DeJong, 1995). However a rigorous mathematical model of the diffusive regime for bounded elastic media lacks at present. It should be able to predict the equilibration ratios as function of the sub-systems mechanical parameters. Another open issue is the consideration of the glancing set of energy rays–the diffractive and gliding ones–for the part of the power flows possibly trapped by the interfaces and junctions. Both aspects are the subject of ongoing research.

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